

DEM 601 Introduction to Demographic Analysis

Exercise Set 2

TERM
Fall 2026

SET IN
lecture 2

WORTH
2% of the final grade

DUE
6 October 2026, 23:59

Readings for this lecture

Rowland, Donald T. (2003) *Demographic Methods and Concepts*
Population Change

Hinde, Andrew (1998) *Demographic Methods*
Some Demographic Fundamentals

Wachter, Kenneth W. (2014) *Essential Demographic Methods*
Periods and Cohorts

Carmichael, Gordon A. (2016) *Fundamentals of Demographic Analysis: Concepts, Measures and Methods*
Basic Sources, Concepts, Definitions and Types of Measures

How to hand this in. By 6 October 2026, 23:59. Show your working: an answer with no arithmetic behind it earns no marks even when the number is right. State the source and the year of every figure you look up yourself. Late work receives no credit.

Every table on this sheet is also a CSV in the `data/exercise02` folder handed out with it, so no figure needs typing in by hand.

1. Your home province, in rates

20 marks

Using TurkStat Vital Statistics, find the registered **births** and **deaths** for your home province in 2015 and in 2025. If you were born outside Türkiye, use İstanbul.

- State the four figures and the province's mid-year population for each of the two years, with the source and the table you took each from, and say which population figure you used and why.
- Compute, for both years and per thousand: the crude birth rate, the crude death rate, and the rate of natural increase.
- Which changed more between 2015 and 2025, the **counts** or the **rates**? Show one sentence of evidence, and say what the counts alone would have led you to believe.

Why not just the counts? This is the lecture 2 opening argument applied to your own province. A count answers "how many". It cannot answer "how fast", or "compared with what".

2. What kind of measure is each one?

24 marks

For each of the following, say whether it is a **ratio**, a **proportion**, a **percentage**, a **rate**, a **probability** or a **mean**. For each one also state **what is in the denominator**, and whether the numerator is inside it.

- a. The number of persons in an age group at mid-year who are employed, divided by the total number of persons in the same age group at mid-year.
- b. The number of neonatal deaths in years x and $x + 1$ occurring to births in year x , divided by the number of births in year x .
- c. The number of unemployed persons who found jobs during a year, divided by the total number of unemployed persons at mid-year.
- d. The number of females with blue eyes, divided by the total number of males and females.
- e. The number of divorces in year x , divided by the number of marriages in year x .

In one of these, the numerator and the denominator do not cover the same people. Identify it and say which of the two figures you would change.

3. Tongoland, 1980

22 marks

The population of Tongoland was **52,000** on 1 January 1980. During 1980:

- 1,200 migrants arrived from a neighbouring country on **1 March** and stayed until the end of the year;
- no other migration was observed;
- births and deaths fell as follows.

Month	Births	Deaths
January	90	50
February	95	60
March	100	30
April	100	30
May	110	30
June	115	30
July	120	20
August	120	20
September	120	20
October	130	30
November	110	50
December	90	50

Assume all months are of equal length, that births and deaths were spread evenly within each month, and that migration, births and deaths were independent.

- Compute the population on 31 December 1980.
- Compute the mid-year population and the total person-years lived. Say which of the two is larger, and what feature of this particular year makes it so.
- Sketch the year on a **Lexis diagram**, with the arrivals on it. Mark the part of the diagram your person-years figure measures.

4. A rate is not a probability

18 marks

The Simeranya cohort from the lecture is followed from exact age 15 to exact age 20. There are **350,056** people alive at age 15 and **342,573** alive at age 20.

- a. How many died between the two ages, and what is the probability that a person alive at 15 dies before 20? Give the probability of surviving as well, and say why the two must sum to 1.
- b. Assuming the deaths were spread evenly over the five years, compute the person-years lived by this cohort between ages 15 and 20, and the death **rate** per person-year over the interval.
- c. Your rate is not equal to your probability divided by five, although the two are close. Explain the difference in terms of what sits in each denominator.

5. The country's own median

16 marks

Age group	Population	Age group	Population	Age group	Population
0-4	4,856,560	35-39	6,275,200	70-74	2,659,275
5-9	6,155,912	40-44	6,432,184	75-79	1,801,528
10-14	6,518,251	45-49	6,194,133	80-84	1,001,898
15-19	6,439,653	50-54	5,506,148	85-89	505,876
20-24	6,268,695	55-59	4,500,240	90+	243,713
25-29	6,620,389	60-64	4,426,895		
30-34	6,314,849	65-69	3,370,769		

TurkStat, ABPRS, 2025.

- Compute the **median age**, showing the interpolation and naming the median group.
- Compute the **mean age**, treating every group as its midpoint. State what you assumed about the open group, say what that assumption is worth, and say which of the two measures is larger and what that says about the shape.

THE POPULATION BLUEPRINT • CHAPTER 2

You confirmed your population in Chapter 1. Now measure it.

1. Find the total population of your chosen population for the most recent year available, and its distribution by five-year age group. State the source, the year, and whether the figures are a census, a register or an estimate.
2. From that distribution, compute **one ratio, one proportion and one percentage**. For each, name the numerator, the denominator, the period and the constant, and say whether the numerator sits inside the denominator.
3. Compute the median age of your population from the grouped distribution, using the interpolation formula from the lecture. Show the working.
4. Write one paragraph on what those four numbers already tell you about the population, and one sentence on what you cannot yet say and why.

If your population has no published age distribution, tell me before the due date rather than after it. That is itself a finding, and we will choose a different level of geography together.

Total: 100 marks